

Engine Only Operates in Idle Mode

Symptoms

- The engine will **only** operate in idle mode.

How To Use This Tree

This symptom tree can be used to troubleshoot engine idle symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Customer Interface Box Wiring	
	STEP 1A. Check Idle/Rated Switch Signal Wire for Short Circuit	Less than 10 ohms resistance?
STEP 2.	Check Engine Harness to Customer Interface Box Cable	
	STEP 2A. Check Idle/Rated Switch Signal Wire	Less than 10 ohms resistance?

STEP 1. Check Customer Interface Box Wiring

STEP 1A. Check Idle/Rated Switch Signal Wire for Short Circuit

Conditions : • Open the customer interface box • Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.		
Action	Specification/Repair	Next Step

Check idle/rated switch signal wire. • Disconnect the idle/rated switch signal wire from the X4 connector. • Place one test lead on the idle/rated switch signal pin in connector C3. • Place the other test lead on the battery 1 voltage return terminal of the customer interface box logic unit.	Less than 10 ohms resistance? YES Repair : Replace the faulty wire(s). Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.
	Less than 10 ohms resistance? NO	2A

STEP 2. Check Engine Harness to Customer Interface Box Cable

STEP 2A. Check Idle/Rated Switch Signal Wire

Conditions :

- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.
- Disconnect customer interface box to engine harness cable connector C10 from the engine harness.

Action	Specification/Repair	Next Step

Check idle/rated switch signal wire. - Place one test lead in the idle/rated switch signal pin of the C3 connector. - Place the other test lead on another pin in the C3 connector. - Repeat for all other pins in the C3 connector.	Less than 10 ohms resistance? YES Repair : Replace the cable.	Repair complete.
	Less than 10 ohms resistance? NO Repair : Refer to OEM Service Manual for idle switch repair instructions.	Repair complete.